

Revision Answers Week 5

Question 1: Briefly explain within the context of Speech Recognition (SR) the following term:

- Speaker Dependent SR
 - **SD-SR** uses the sounds of a **single** speaker to learn the parameters or models that characterise the system's internal model of the speech process. Thus the system is used to recognize the speech of its trainer.
- Speaker Independent SR systems
 - **SI-SR** systems are trained by **multiple** speakers and used to recognize even speakers outside the training population.
 - For the same level of complexity, **SD** systems are generally more accurate than **SI** systems.

Question 2: Briefly explain the following terms:

- Speaker identification
- Speaker verification

Please see the lecture slide on
SR Use Cases: Identification and Verification

Question 3: Briefly describe the role of Endpoint detection in a SR system.

- Detects the presence of speech in a background of noise. Also, accurately detect the beginning and the end of an utterance.
- The task of separating speech from background "silence" is **trivial** only in acoustic environments with extremely **high signal-to-noise ratios** (SNRs).

Question 4: Provide a functional block diagram of a Generic Speech Production and Perception system.

Question 5: The level of difficulty in achieving high speech recognition performance and system complexity

- Increases exponentially with vocabulary size
- Increases logarithmically with vocabulary size
- Increases linearly with vocabulary size
- Remains constant with vocabulary size
- None of the above

Briefly explain your answer.